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Homework 5

Graduate Requirement Research

Different approaches exist for the problem of training models to detect Polyp-Segmentation in Colonoscopy Images. The need for this is obvious. Identifying polyps is important for identifying and treating cancer. It is no surprise then that there are many intelligent individuals working on developing accurate models that can improve the accuracy and efficiency of the screening process. I will discuss a few of them below.

The first model I will showcase is the Mask R-CNN object detection neural network. Specifically, it is an ensemble method that combines two Mask R-CNN (Region-based Convolutional Neural Network) models with different backbone structures such as ResNet50 and ResNet101. It consists of three components: data augmentation, two Mask R-CNNs with different backbone structures, and the bitwise combination of two masks to enhance the segmentation performance. Data augmentation was implemented, consisting of vertical flipping, horizontal flipping, random rotations and more. According to "Ensemble of Instance Segmentation Models for Polyp Segmentation in Colonoscopy Images" these help to make up for a deficiency in the quantity of available images and help to reduce overfitting.

This model showed strong results that surpassed state of the art models on challenging benchmarks like COCO for detection, segmentation, and key-point tasks. Unfortunately, there are some disadvantages to this model as it is computationally taxing, requires a very large amount of already labeled images, and requires a long processing time. However, it was worth discussing as its ensemble method provides a unique solution to overcoming high-dimensional, non-convex loss functions with many local minima by using, which the paper claims makes it "virtually guaranteed to converge to different solutions....".

Further it is also worth assessing their evaluation metrics. The team uses three datasets to evaluate their methods and compared the results to state of the art models. Three common

segmentation evaluation metrics were evaluated: mean pixel precision, mean pixel recall and intersection over union. They are calculated using false positives (FP) which in this case are every pixel segmented as polyp that falls outside of a polyp mask and true positives which are every pixel of a polyp that is correctly classified. The equations can be seen in this image included in the paper.

$$\begin{aligned} PR &= \frac{TP}{TP + FP} \\ RC &= \frac{TP}{TP + FN} \\ IU &= \frac{TP}{TP + FP + FN} \end{aligned}$$

While the results of this model were promising, the research is not yet complete and the barrier to entry on this type of model is high due to the need for the large amount of already classified images and significant preprocessing. Therefore, we move on to the next model.

The next model we discuss is U-Net. It is a convolutional network architecture comprised of an encoder and a decoder structure. It utilizes skip connections and is capable of giving the “precise delineation of object boundaries...” (TS, Chandrakantha..., P.3). An article published in the International Journal of Scientific Research in Science and Technology found that U-Net demonstrated exceptional performance in polyp segmentation. In this study, the metric used was Mean IoU (intersection over union). In short, this measures the overlap between the predicted polyp masks and the given “ground truth” polyp masks. This paper found that U-Net achieved a Mean IoU score of 0.9897. For comparison the same study found that a different model, DeepLab displayed a slightly less impressive IoU score of 0.9676.

Therefore, for the purposes of this project. I chose to go with the U-Net model for its promising results as well as its feasibility. The specific version of the polyp database I used was [Angelou0516/CVC-ClinicDB](#) from Hugging face. The results I got were not as good. The best IoU score I got was 0.8106, even after implementing tricks I found from reading these papers,

such as BCE Dice loss, image augmentations, and tuning the hyperparameters. However, I consider these results to be promising. Perhaps more convolutional layers would yield better results. A different variation of U-Net may also yield better results.

Works Cited:

J. Kang and J. Gwak, "Ensemble of Instance Segmentation Models for Polyp Segmentation in Colonoscopy Images," in *IEEE Access*, vol. 7, pp. 26440-26447, 2019, doi: 10.1109/ACCESS.2019.2900672.

TS, Chandrakantha, and Basavaraj N. Jagadale. "Comparative analysis of u-net and deeplab for automatic polyp segmentation in colonoscopic frames using cvc-clinicdb dataset." *J. Med. Imaging Health Inform* 13 (2023): 1-10.